

Preparation and characterization of melt-stretched polypropylene-polypropylene-*g*-poly(α -methyl styrene-*co*-glycidyl methacrylate-*co*- γ -methacryloxypropyl trimethoxy silane)-silicon dioxide compound microporous membranes

Shan Jiang , Wei Wang, Wei Chen, Hongwen Zhang, Kemin Wang

School of Materials Science and Engineering, Changzhou University, Changzhou 213164, Jiangsu, China

Correspondence to: S. Jiang (E-mail: jiangshan@cczu.edu.cn)

ABSTRACT: Polypropylene (PP)-silicon dioxide (SiO₂) compound microporous membranes were fabricated by a melt-stretching method. Although the permeability, porosity, and hydrophilicity values of the microporous membranes were found to be highest at an SiO₂ content of 2 wt %, the heat resistance of the membranes was relatively low. To improve the heat resistance of the microporous membranes, a macromolecular coupling agent, PP-*g*-poly(α -methyl styrene-*co*-glycidyl methacrylate-*co*- γ -methacryloxypropyl trimethoxy silane) (PAGK), was introduced into the membrane. In the PP-PP-*g*-PAGK-SiO₂ composite systems, the content of SiO₂ was controlled at 2 wt %, and the proportion of PP-*g*-PAGK was varied. With increasing PP-*g*-PAGK content to 0.6%, the Gurley value decreased from 250 to 239 s, and the porosity increased from 50.8 to 51.6%. The hydrophilicity of the microporous membranes increased with the incorporation of PP-*g*-PAGK, and their water vapor transmission rate reached a maximum of 3360 g m⁻²/24 h at a PP-*g*-PAGK content of 0.6%. The heat resistance of the PP-PP-*g*-PAGK-SiO₂ compound microporous membranes was slightly higher than that of the pure PP microporous membrane. Additionally, the cycle performance of a cell assembled with the PP-PP-*g*-PAGK-SiO₂ membrane was better than that constructed with the pure PP membrane. © 2019 Wiley Periodicals, Inc. *J. Appl. Polym. Sci.* **2019**, 136, 47937.

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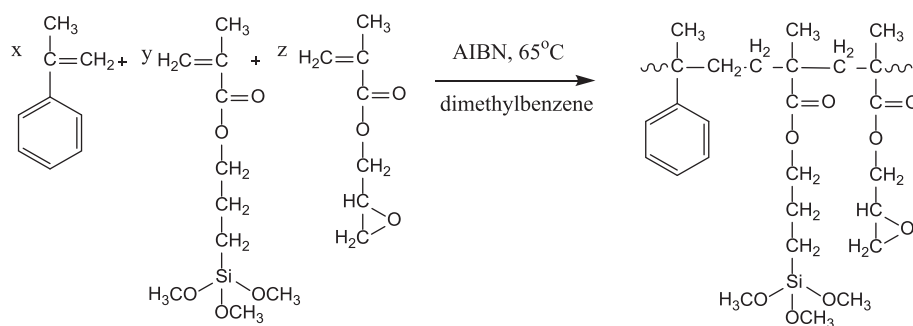
INTRODUCTION

Polyethylene and polypropylene (PP)-based polyolefin microporous membranes are widely used as lithium-ion battery separators.^{1,2} PP separators have an excellent chemical stability and mechanical strength and a low manufacturing cost. Although the separators currently being used can meet daily needs, there still exist two main drawbacks: a poor electrolyte wettability and poor heat resistance. Because of their nonpolarity and low surface energy, PP separators exhibit a poor affinity with traditional electrolytes; this hinders the passage of polar electrolytes through micropores.³ When the internal circuit of a battery is short or a high current density is used for a long time, the high internal temperature generated leads to shrinkage in the polymer film. Such excessive heat accumulation often leads to safety hazards associated with lithium-ion batteries;⁴ this is of great significance for improving the heat resistance and surface polarity of separators.

To counter the disadvantages of poor electrolyte wettability and heat resistance, surface modification and blending methods are often used. Yu and colleagues used N₂,⁵ NH₃,⁶ O₂,⁷ and air⁸ separately as plasma treatment reagents to modify PP microporous

membranes and studied the effects of the plasma treatment time and power on the micromorphology, hydrophilicity, and water flux of the membranes. Silicon dioxide (SiO₂), TiO₂, and Al₂O₃ particles were introduced onto the surfaces of microporous membranes with binders, such as poly(vinylidene fluoride). Saffar *et al.*⁹ prepared hydrophilic PP microporous membranes by blending PP with hydrophilic maleic anhydride or acrylic acid grafted PP. The effects of these modifiers on the properties of the precursor membranes and microporous membranes were studied. We found that the hydrophilicity of the modified microporous membranes was higher compared to that of unmodified membranes.

In this study, we used SiO₂ to improve the electrolyte wettability and heat resistance of separators, and a macromolecular coupling agent, PP-*g*-poly(α -methyl styrene-*co*-glycidyl methacrylate-*co*- γ -methacryloxypropyl trimethoxy silane) (PAGK) was used to improve the dispersion of SiO₂ in PP. In our previous studies, a copolymer prepared by the copolymerization of α -methyl styrene (AMS) and glycidyl methacrylate (GMA) decomposed into macroradicals when the temperature was above 80°C. The AMS units were supposed to be weak and responsible for the thermal decomposition. When this



Scheme 1. Synthesis reaction of PAGK.

special copolymer was added to a PP substrate at its melting temperature, the copolymer acted as a macroinitiator and made the AMS–GMA chains graft onto PP chains without extra monomers or initiators.^{10–12} Keeping this in mind, we designed and synthesized the copolymer PAGK via the copolymerization of AMS and the functional monomers GMA and γ -methacryloxypropyl trimethoxy silane (KH570) and later grafted PAGK onto PP chains by melt blending. Microporous membranes were prepared by a melt-stretching method, and their properties were evaluated.

EXPERIMENTAL

Materials

A homo-PP resin (F401) was purchased from Liaoning Huajin Chemical Industry Group Co., Ltd. The melt flow rate was 2.4 g/10 min (under ASTM D 1238 conditions of 230°C and 2.16 kg). SiO₂ was purchased from Degussa (A200) and had a diameter of 12 nm. AMS and GMA were purchased from Aladdin. Azobisisobutyronitrile (AIBN) was purchased from Shanghai ShiSiHeWei Chemical Co., Ltd., and recrystallized from EtOH. KH570 and other reagents were purchased from Sinopharm Chemical Reagent Beijing Co., Ltd.

Synthesis of the Macroinitiator PAGK

The monomers [AMS (0.067 mol), GMA (0.53 mol), and KH570 (0.067 mol)], initiator [AIBN (0.0446 mol)], and solvent [dimethyl benzene (250 mL)] were added to a three-necked flask, and N₂ was bubbled through the mixture for 30 min. This particular molar ratio was chosen for its high yield. The flask was then placed in an oil bath. The polymerization was carried out for 24 h under an N₂ atmosphere at 65°C. After it was cooled to room temperature, the reaction mixture was precipitated and then washed three times with an excess amount of methanol. Finally, the polymer was dried in a vacuum oven at 65°C for 24 h. The product obtained exhibited a yield of 64.3% and was stored for later use. The number-average molecular weight of PAGK was 15 400, whereas its molecular weight distribution (weight-average molecular weight/number-average molecular weight) was 1.87 (Scheme 1).

Preparation of PP-g-PAGK

The macromolecular coupling agent, PP-g-PAGK, was obtained by the melt blending of PP with PAGK. The grafting reaction of PP with the PAGK macroinitiator was carried out in a corotating intermeshing twin-screw extruder from Nanjing Guangda. PP (1000 g) and PAGK (50 g) were introduced into the hopper of the extruder after they were fully mixed in a high-speed disperser. Extrusion was carried out at a screw rotation speed of 90 rpm, and the grafting

temperature was 220°C. The extrudates were cooled in a water bath, pelletized, and then dried at 60°C for 8 h. The extrudates were dissolved in dimethyl benzene at 140°C to remove residual PAGK and then rapidly precipitated with an excess of acetone. Finally, a white powder was obtained after extraction and drying.

The PP-g-PAGK powder (before and after purification) was pressed into a 100–120 μ m thick film at 230°C with a flat vulcanizer, and the grafting efficiency and grafting rate of PP-g-PAGK were measured by IR spectroscopy. The absorption peak of PP-g-PAGK at 1727 cm⁻¹ was attributed to the C=O groups in the GMA and KH570 units. The absorption peak at 2724 cm⁻¹ represented the sensitive peak of PP, and the following equations were used for these calculations:

$$c = \frac{A_{1727}/A_{1727}}{A_{1727}/A_{2724}} \quad (1)$$

$$g = c \times w\% \quad (2)$$

where c is the grafting efficiency of PAGK; g is the grafting ratio of PP-g-PAGK; A_{1727} and A_{2724} are the areas of the sensitive characteristic peaks of PAGK and PP in the unpurified PP-g-PAGK, respectively; A_{1727} and A_{2724} are the areas of the sensitive characteristic peaks of PAGK and PP in the purified PP-g-PAGK, respectively; and $w\%$ is the weight fraction of PAGK relative to PP in raw material.

Preparation of the PP–SiO₂ and PP–PP-g-PAGK–SiO₂ Composite Particles

PP (700 g) and SiO₂ (0, 0.5, 1, 2, 4, 6, and 8 wt %) were fully mixed in a high-speed disperser, and PP–SiO₂ composite particles were prepared with a twin-screw extruder.

To prepare the PP–PP-g-PAGK–SiO₂ composite particles, 700 g of PP, 14 g of SiO₂, and PP-g-PAGK (0, 0.6, 1.8, 4, and 6 wt %) were fully mixed in a high-speed disperser, and we prepared the composite particles by extruding this mixture on a twin-screw extruder.

Preparation of the Precursor Films and Stretched Microporous Membranes

The precursor films of PP–SiO₂ and PP–PP-g-PAGK–SiO₂ were prepared by cast extrusion, and then, they were annealed at 145°C for 30 min in a hot oven. The annealed films were first stretched to 15% at room temperature, then stretched to 100% at 145°C, and gave birth to the microporous membranes.¹²

Electrode Preparation and Cell Assembly

A LiFePO₄ positive electrode (Super-P) was prepared by the coating of an *N*-methyl pyrrolidone based slurry of dried LiFePO₄,

poly(vinylidene fluoride), and carbon at a weight ratio of 8:1:1 on aluminum foil. After vacuum drying, the electrode was punched into a disk. A lithium plate was used as the negative electrode and was also punched into a disk. $\text{LiFePO}_4/\text{Li}$ half-cells were then assembled via the sandwiching of the prepared PP-based separators between the Li negative electrode and LiFePO_4 positive electrode. The cells were then enclosed in a 2025 coin cell and injected with commercial LBC305-01 electrolyte solution. All of the cells were assembled in a dry box filled with argon gas.

Characterization

Fourier Transform Infrared (FTIR) Spectroscopy. An FTIR spectrometer (Nicolet) was used to characterize the films. The orientation degree of the samples was also characterized by FTIR spectroscopy, and the crystalline phase (f_c) was calculated. The beam was polarized by means of a Spectra-Tech zinc selenite wire grid polarizer. The dichroic ratio (D) was calculated from the ratio of the absorbance from beams polarized parallel to the melt extrusion direction ($A_{||}$) to the absorbance from beams polarized perpendicular to the melt extrusion direction ($A_{\perp}$). Then, the data were evaluated with the degree of orientation with eq. (3):

$$D = \frac{A_{||}}{A_{\perp}} \quad (3)$$

For the PP membranes, the absorption at a wave number of 998 cm^{-1} was attributed to f_c , and the orientation of f_c was obtained by eq. (4):

$$f_c = \frac{D_{(998)} - 1}{D_{(998)} + 2} \quad (4)$$

where $D_{(998)}$ is the dichroic ratio value for the 998 cm^{-1} band.

$^1\text{H-NMR}$ Spectroscopy. The $^1\text{H-NMR}$ spectra were recorded with a Bruker Avance ARX-400 instrument.¹²

Gel Permeation Chromatography. The molecular weight and molecular weight distribution of PAGK were determined with gel permeation chromatography (Waters 515 system) with tetrahydrofuran as an eluent.

Differential Scanning Calorimetry. The degrees of crystallinity (X_c) of the cast films were evaluated on a DSC-8000 instrument.¹²

Water Contact Angle. The water contact angle was determined with a contact angle apparatus (Beijing Hake Experiment Instrument Factory).¹²

Elastic Recovery (E_R). The E_R values of the cast films were analyzed with an electromechanical universal testing machine from MTS Systems Co., Ltd., at a rate of 50 mm/min . E_R was determined at room temperature along the machine direction, and the percentage E_R was calculated with eq. (5):

$$ER = \frac{L - L'}{L - L_0} \times 100\% \quad (5)$$

where L_0 is the initial length of the precursor films before extension, L is the length after stretching, and L' is the length at the end of extension.

Porosity and Air Permeability of the Microporous Membranes. The porosity of the fabricated membranes was measured by

imbibition. The air permeability of the microporous membranes was characterized with a Gurley densometer (model 4110, Gurley Precision Instruments, New York).

Scanning Electron Microscopy (SEM). The surface morphology was observed with a field emission scanning electron microscope (SUPRA55, Zeiss, Germany).¹² The surface porosity (P_s) and bridge length (l_b) of each sample was obtained via analysis of the corresponding images with an image processing software called Image-Pro Plus.¹³

Water Vapor Transmission Rate (WVTR). The water vapor permeability of the membranes was measured with a GB/T 30412-2013 instrument. Sample bottles filled with deionized water were sealed with the microporous membranes. The bottles were all placed in a constant-temperature, constant-humidity box maintained at a relative humidity of 25% and 25°C . The bottles were kept in the box for 24 h, and their weights before and after weight loss were accurately measured. The arithmetic average of five sets of data was used to calculate WVTR.

Electrochemical Performance. Charge-discharge tests on the assembled cells were conducted in the voltage range of $2.5\text{--}4.2\text{ V}$ with a Neware system at room temperature. To protect the cells, the first charge-discharge process was conducted at a C rate, a measure of the rate at which a battery is discharged relative to its maximum capacity, of 0.1 and the cycle performance tests were conducted at a current density of 0.5 C. The C-rate capability of the cells was conducted at various current densities in the range of $0.1\text{--}10\text{ C}$.

RESULTS AND DISCUSSION

Characterization of the PP-SiO₂ Compound Microporous Membranes

The lamellar orientation of the precursor films affects the pore-formation properties of melt-stretched PP microporous membranes. In general, the orientation of the lamellar structures of precursor films is usually characterized in terms of their X_c , crystalline orientation parameter, and E_R . Additionally, annealing is important in the preparation of microporous membranes. The annealing of precursor films at a certain temperature allows the rearrangement of lamellar crystals along the orientation direction; this enhances the crystallization orientation and reduces defects in the precursor films.

Figure 1 illustrates the changes in X_c and the crystalline orientation parameter of the PP-SiO₂ precursors and annealed films with SiO₂ content. Figure 1(a) indicates that the X_c values of the films before and after annealing were about 45 and 49.5%, respectively. Annealing increased the X_c of the precursor films, and the changes in the trend of X_c of the film before and after annealing were the same. Before annealing, the X_c of the pure PP film was 44.67%. When the content of SiO₂ was 2 wt %, the X_c of the precursor film achieved a maximum of 45.92%. SiO₂ particles could have acted as nucleating agents in the composite basement film. When the SiO₂ contents were 6 and 8 wt %, the X_c s of the precursor film were equivalent to that of the pure PP film. Annealing increased the X_c of the pure PP film by 5.12%, and the X_c of the film containing 2 wt % SiO₂ increased by 4.2%. However, when

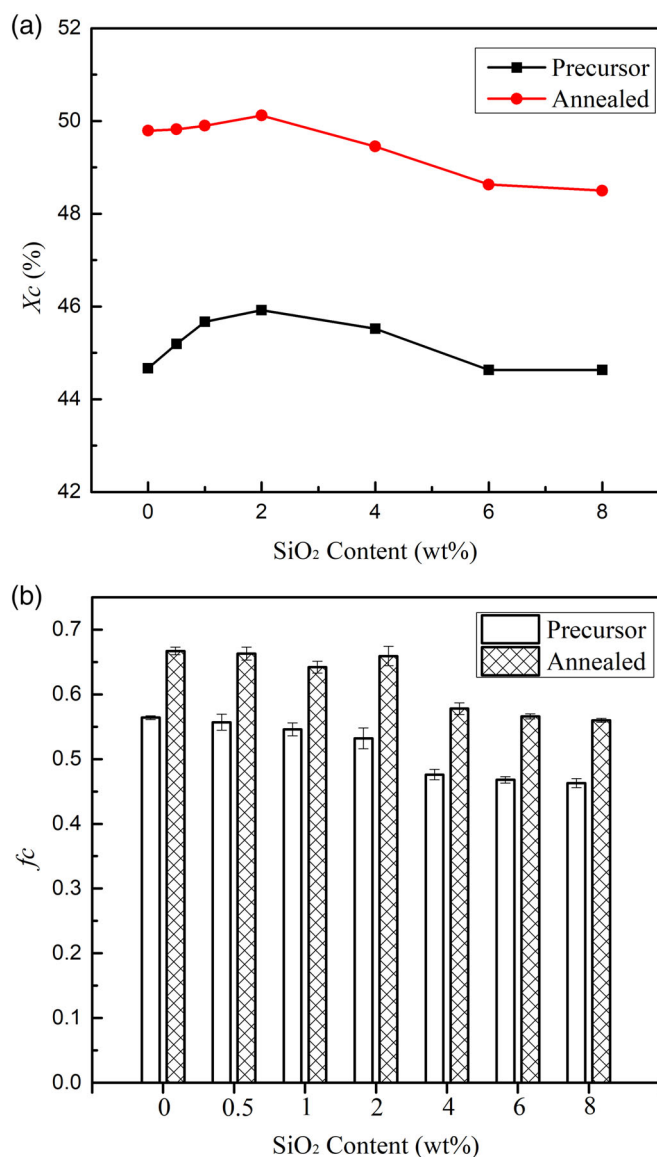


Figure 1. (a) X_c and (b) f_c of the PP- SiO_2 precursor and annealed films with different SiO_2 contents. [Color figure can be viewed at wileyonlinelibrary.com]

the SiO_2 content was 8 wt %, the X_c of the film increased by only 3.97%.

Figure 1(b) shows an analysis of the changes in the crystalline orientation parameters of the PP- SiO_2 precursor and annealed films at different SiO_2 contents. The introduction of SiO_2 particles degraded the crystalline orientation of the precursor films, and the crystalline orientation parameter decreased more prominently beyond 2 wt % SiO_2 . As annealing was performed at a temperature close to the onset temperature of mobility in the crystalline structure, we postulated that during annealing, the lamellar twist

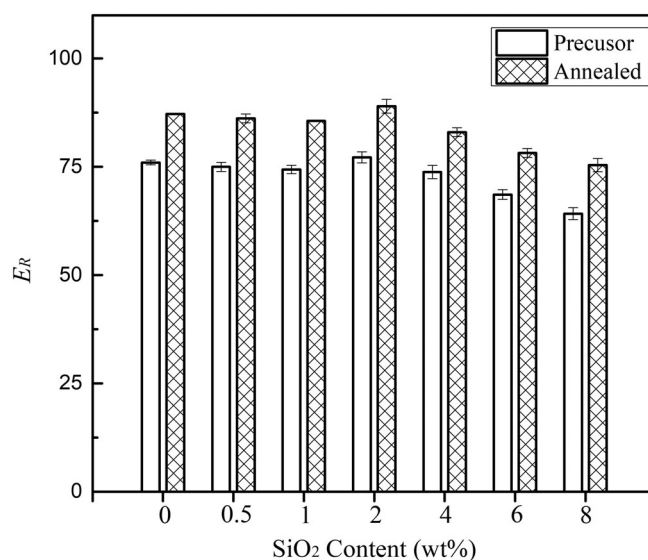


Figure 2. E_R values of the PP- SiO_2 precursor and annealed films with different SiO_2 contents.

and orientation were perpendicular to the machine direction. Furthermore, the melting of small lamellae occurred; this led to their recrystallization with a better orientation.¹⁴ After annealing, the crystalline orientation parameter of the films improved noticeably. In particular, in the sample with 2 wt % SiO_2 , the crystalline orientation parameter increased greatly after annealing; this was ascribed to the nucleating effect of SiO_2 during annealing.

E_R can be used to characterize the elastic deformation properties of the lamellar structure of precursor films; it is largely determined by the orientation of the lamellar structure of the precursor films. Figure 2 shows the variation in the E_R values of the precursor and annealed films with respect to the SiO_2 content. We observed that when the content of SiO_2 was varied in the range 0.5–2 wt %, the E_R of the precursor film changed little with respect to the pure PP sample, although there was a slight increase at 2 wt %. However, excessive SiO_2 reduced both X_c and the crystalline orientation of precursor film. When the SiO_2 content reached 6 wt %, the crystalline orientation of the precursor films was destroyed, and E_R was reduced. Hence, with respect to the SiO_2 content, the rate of E_R of the annealed films was similar to that of the precursor films.

Changes in the lamellar orientation in the precursor and annealed films affected the pore-forming properties and pore structure of the microporous membranes. The pore structure of the microporous membranes is often indirectly characterized by the Gurley value and porosity, whereas the microstructure of the microporous membranes was observed directly by SEM.

The Gurley values and porosity parameters of microporous membranes with different SiO_2 contents are given in Table I. In this

Table I. Gurley Values and Porosities of PP- SiO_2 Microporous Membranes with Different SiO_2 Contents

SiO_2 content (wt %)	0	0.5	1	2	4	6	8
Gurley value (s)	260	255	254	250	283	289	290
Porosity (%)	49.2	49.3	49.6	50.8	48.3	47.8	47.2

study, the Gurley value indicated the time taken by 100 mL of air to pass through a 645.2 mm^2 microporous membrane. An increase in the Gurley value indicated a decrease in the microporous membrane permeability. On the other hand, the permeability of the membrane improved. The Gurley value of the microporous membranes decreased slightly when the SiO_2 content increased from 0 to 2 wt %; the permeability of the microporous membrane was highest when the SiO_2 content was 2 wt %. With a further increase in the SiO_2 content, the Gurley value also increased; this indicated permeability reduction, which may have been caused by a poor pore structure induced by the aggregation of SiO_2 particles.

The porosity of the microporous membranes was measured by the imbibition method; an increase in the porosity indicated an increase in the proportion of pores in the microporous membranes. When the SiO_2 content was lower than 2 wt %, the porosity of the microporous membranes was slightly greater, and the porosity of the microporous membrane was as high as 50.8% when the SiO_2 content was 2 wt %. However, it decreased when the SiO_2 content increased beyond this value; this may have been due to a reduction in the pore-forming ability of the precursor films.

Figure 3 shows the SEM images of microporous membranes with different SiO_2 contents. No obvious SiO_2 particles were seen in the images in Figure 3; this indicated that the inorganic SiO_2 particles were wrapped in PP. Typical connecting bridges and pores associated with melt-stretched microporous membranes were clearly observed; these were formed because of the separation of the lamellar structure rather than phase separation between the inorganic SiO_2 particles and PP polymer. The pore sizes of the microporous membranes with SiO_2 contents in the range 0.5–2 wt % [Figure 3(b–d)] were larger than those in the pure PP microporous membrane [Figure 3(a)]. However, with a further increase in the SiO_2 content [Figure 3(e,f)], the pore size of the microporous membrane decreased, and the micropores became denser; meanwhile, a large number of broken connecting bridges appeared. We found that a small amount of agglomerated SiO_2 particles (in red circles) appeared [Figure 3(e–g)]; agglomeration was highest in Figure 3(g). Those agglomerated SiO_2 particles were distributed between the unopened main lamellae. Combined with X_c and E_R of the precursor, we deduced that excessive SiO_2 particles agglomerated and reduced both the lamellar content and porosity.

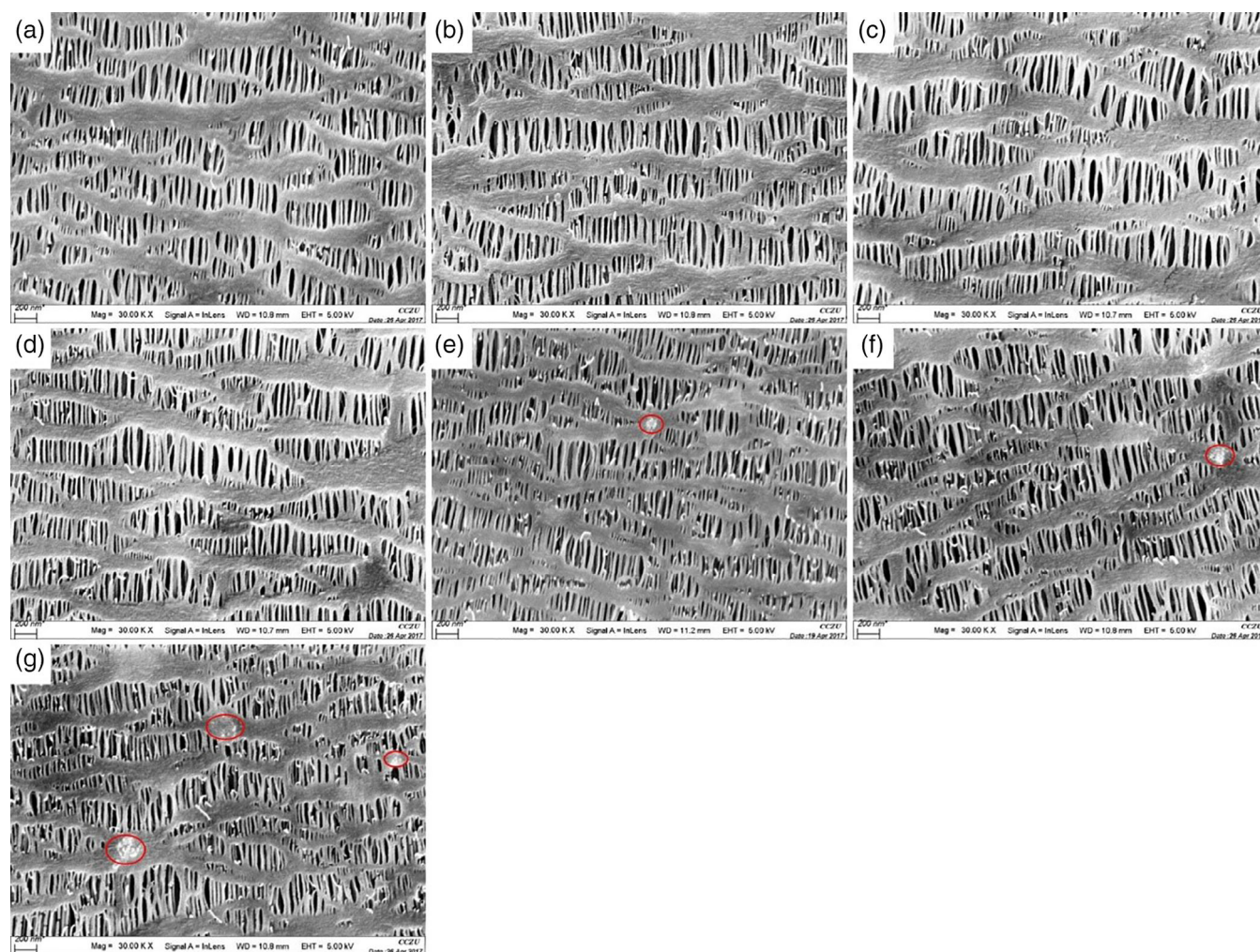


Figure 3. SEM images of PP- SiO_2 microporous membranes with SiO_2 contents of (a) 0, (b) 0.5, (c) 1, (d) 2, (e) 4, (f) 6, and (g) 8 wt %. [Color figure can be viewed at wileyonlinelibrary.com]

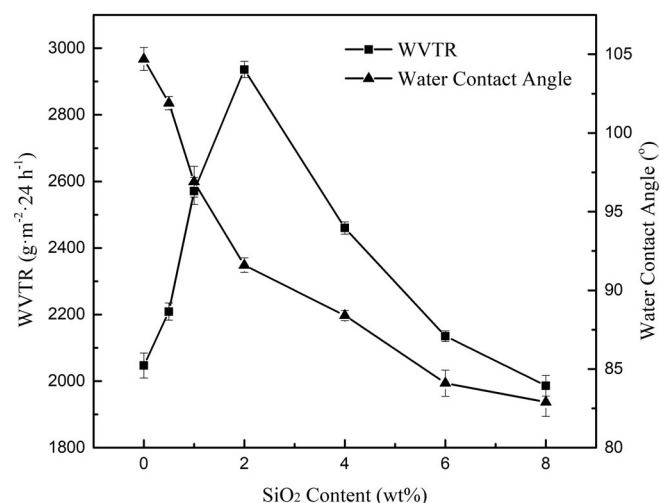


Figure 4. WVTRs and water contact angles of PP-SiO₂ microporous membranes with different SiO₂ contents.

The electrolyte commonly used in lithium batteries is a polar solution. However, traditional PP is a typical nonpolar polymer; hence, traditional PP microporous membranes cannot be wetted very well by electrolytes. When hydrophilic SiO₂ is introduced, changes in the hydrophilicity can be used indirectly to characterize the polarity of microporous membranes.

Figure 4 shows the WVTR and water contact angle results of PP-SiO₂ microporous membranes with different SiO₂ contents. We observed from the plot that with an increase in the SiO₂ content in PP, the water contact angle of the microporous membranes gradually decreased. However, because most of the SiO₂ particles were embedded in PP (refer to the SEM images), there was limited reduction in the water contact angle; it decreased from 104.7 to 82.9°.

As shown in Figure 4, in the SiO₂ content range of 0–2 wt %, the WVTR of the microporous membranes increased gradually and reached a maximum value of 2936 g m⁻²/24 h when the SiO₂ content was 2 wt %. When the SiO₂ content was higher than 2 wt %, the WVTR began to decrease. The WVTR at 8 wt % SiO₂ was 1986 g m⁻²/24 h; this was equivalent to that of the pure PP microporous membrane. WVTR was affected by the pore structure (porosity and permeability) and the hydrophilicity of the microporous membranes. Previously, we discussed that when the SiO₂ content was 0–2 wt %, the porosity and permeability of the microporous membrane increased gradually but decreased beyond 2 wt %. Therefore, we understood that the microporous membranes provided channels for water vapor to pass through when the SiO₂ content was 2 wt %. At the same time, the hydrophilicity of the microporous membrane also increased, and this improved the WVTR.

To simulate the heating process of a lithium battery, heat-resistance experiments were conducted at 110, 120, 130, 140, and 150°C for 30 min, and an increase in the rate of thermal shrinkage represented a decrease in the heat resistance. Figure 5 depicts the heat shrinkage rate of PP-SiO₂ microporous membranes at different SiO₂ contents. The heat resistance of the microporous membranes decreased with the introduction of SiO₂ particles. On

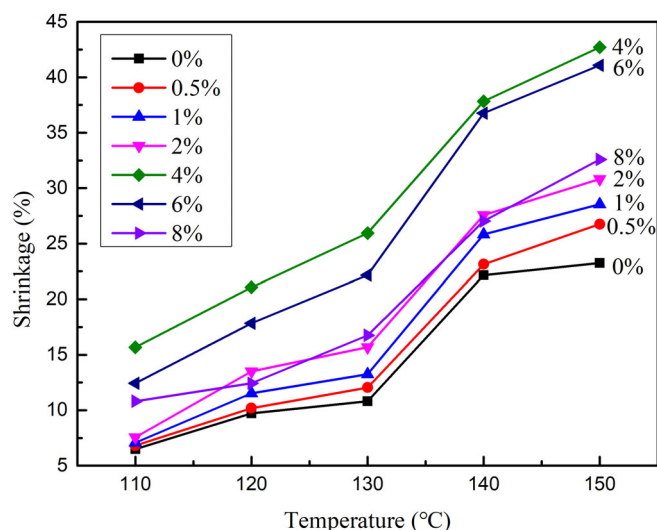


Figure 5. Thermal shrinkage rate of PP-SiO₂ microporous membranes with different SiO₂ contents. [Color figure can be viewed at wileyonlinelibrary.com]

the whole, when the SiO₂ content was lower than 4 wt %, the thermal shrinkage rate of the microporous membrane increased gradually at each temperature, but when the content of SiO₂ was higher than 4 wt %, the thermal shrinkage rate of the microporous membranes began to decrease. The thermal shrinkage rates of 8 and 2 wt % SiO₂ microporous membranes were similar, but they were still higher than that of the pure PP microporous membrane. A possible explanation for the previously described phenomena was that a certain amount of SiO₂ particles aggregated and induced the surrounding molecular chains to contract during the heating process; this resulted in a decrease in the heat resistance. However, when the content of SiO₂ reached a certain level (8 wt %), SiO₂ in the base resin took a larger proportion and distributed in a denser manner. Thus, the heat resistance improved slightly. In conclusion, when the SiO₂ content was 2 wt %, the orientation of the lamellar structures in the precursor film was the best, and the pore structure of the corresponding microporous membrane was perfect. In addition, the hydrophilicity and WVTR were also high, but the heat resistance of the microporous membrane decreased compared to that of the pure PP microporous membrane.

Characterization of the PP-PP-g-PAGK-SiO₂ Compound Microporous Membranes

On the basis of previous research results that showed that when the SiO₂ content was 2 wt %, the orientation of lamellar structures in the precursor film was the best and the pore structure of the corresponding microporous membrane was perfect, the SiO₂ content was kept at 2 wt % in the follow-up experiments. A macromolecular coupling agent, PP-g-PAGK, was synthesized and introduced to improve the dispersion of SiO₂ in PP to improve the heat resistance of the compound microporous membranes. With the addition of PP-g-PAGK, segments of KH570 react with —OH on the surface of silica to improve the compatibility of PP and SiO₂.

Figure 6 shows the ¹H-NMR spectrum of the PAGK macroinitiator. The chemical shifts of benzene-ring hydrogen atoms corresponding to the AMS chains were approximately 7.12 and 7.22 ppm (1–5). The chemical shifts at 3.79 and 4.27 ppm were related to the

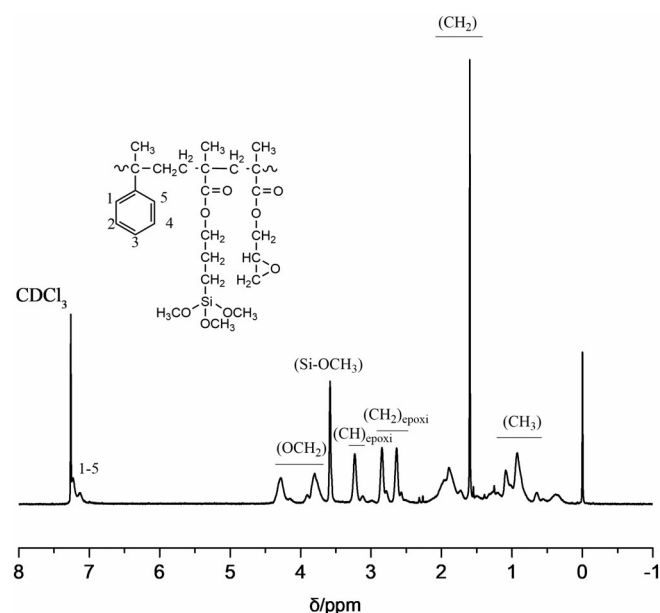


Figure 6. ^1H -NMR of PAGK (with CDCl_3 as the solvent).

characteristic peaks of methylene groups ($-\text{CH}_2-$) connected to the ester bonds in the KH570 and GMA segments, whereas the chemical shifts at 3.83 and 3.63 ppm were the characteristic peaks of $-\text{CH}_2-$ in the GMA units. The chemical shifts corresponding to $-\text{Si}-\text{OCH}_3$ in KH570 were around 3.58 ppm. The chemical shift at 3.24 ppm was attributed to the characteristic peaks of the methenyl ($-\text{CH}-$) groups in the GMA units of the epoxy structure. The AMS/GMA/KH570 molar ratio was calculated to be 1:6.75:0.6 via comparison of the signal intensities of $-\text{CH}_3$ groups in AMS, $-\text{CH}_2-$ groups in the GMA units, and $-\text{Si}-\text{OCH}_3$ groups in the KH570 units.

PP-g-PAGK was then prepared via the melt blending of PP with PAGK; this was followed by purification. During the melt blending, weak bonds existing in the AMS units of PAGK broke into macroradicals, which could further graft onto the PP chains. Figure 7 shows the FTIR spectra of PP, purified PP-g-PAGK, and unpurified PP-g-PAGK. Compared to the pure PP, the purified PP-g-PAGK exhibited a new peak around 1727 cm^{-1} ; this was attributed to the C=O groups in the GMA and KH570 units. In addition, the absorption peak of the unpurified PP-g-PAGK at 1727 cm^{-1} was stronger than that of its purified counterpart. This means that there was a partially nongrafted macromolecular initiator present during melt blending, whereas the process of dissolving and extraction at high temperature could remove the free impurities and purify the product. The grafting efficiency of the graft copolymer was 53.44%, whereas its grafting rate was 2.67%.

PP-g-PAGK was used to improve the dispersity of the SiO_2 nanoparticles and the heat resistance of the microporous membranes. The relationship between X_c of the precursor and annealed films with different contents of PP-g-PAGK is shown in Figure 8(a). The effect of annealing on the precursor films was the same as that in the PP- SiO_2 films. Annealing improved the X_c of the base film, and small imperfect lamellae were further improved under the annealing conditions. Before annealing, the X_c of the precursor film without PP-g-PAGK was 45.92%. When

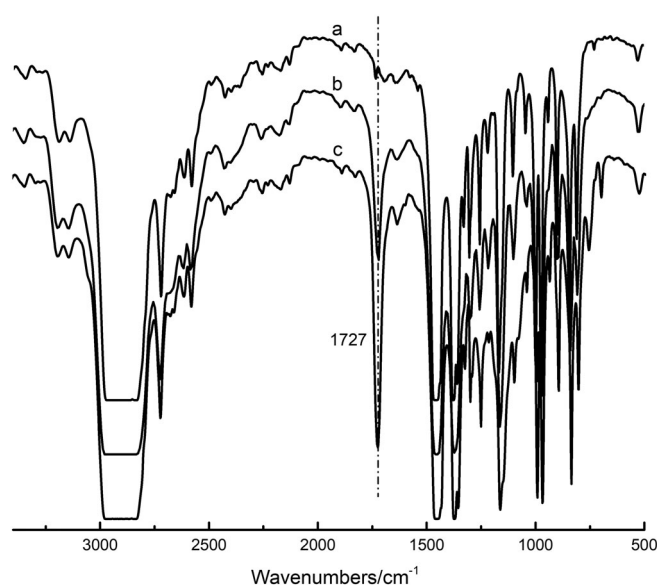


Figure 7. FTIR spectra of the (a) PP, (b) purified PP-g-PAGK, and (c) unpurified PP-g-PAGK.

0.6 wt % PP-g-PAGK was added, the X_c of the film was 46.10%. When the content of PP-g-PAGK was 1.8 wt %, the X_c of the precursor film was around 45.9%.

Figure 8(b) shows the variation in the crystalline orientation parameters of the PP-PP-g-PAGK- SiO_2 precursor and annealed films with PP-g-PAGK content. The introduction of PP-g-PAGK into the system reduced the crystalline orientation parameter of the precursor films. The crystalline orientation parameter of the pure PP precursor film was 0.532, and it decreased to 0.452 when the content of PP-g-PAGK was 6 wt %; this indicated that the introduction of PP-g-PAGK was not conducive to the growth of lamellae along the orientation direction. A similar observation was made with the PP- SiO_2 films; further, annealing increased the crystalline orientation parameter of the PP-PP-g-PAGK- SiO_2 composite films to a great extent.

Figure 9 shows the changes occurring in the E_R rate with respect to the content of the macromolecular coupling agent. We observed that the E_R rate of the pure PP film was 77.2%. When the content of PP-g-PAGK was below 1.8 wt %, the E_R rate of the precursor film varied little. With a further increase in the PP-g-PAGK content, lamellar orientation in the precursor film was destroyed, and the E_R rate was reduced to 65.2% when the PP-g-PAGK content was 6 wt %. Similar to previously discussed results, the E_R rate of the PP-PP-g-PAGK- SiO_2 composite films could be improved by annealing.

In the next part of the study, microporous membranes were obtained by stretching, and their properties were studied. Table II shows the Gurley and porosity parameters of the microporous membranes with different contents of the macromolecular coupling agent. The Gurley value of the microporous membrane without PP-g-PAGK was 250 s; the Gurley value decreased to a large extent upon the addition of 0.6 wt % PP-g-PAGK, but with a continuous increase in the PP-g-PAGK content, the Gurley values increased slightly. The porosity of the microporous membrane without any coupling agent was 50.8%; it reached a

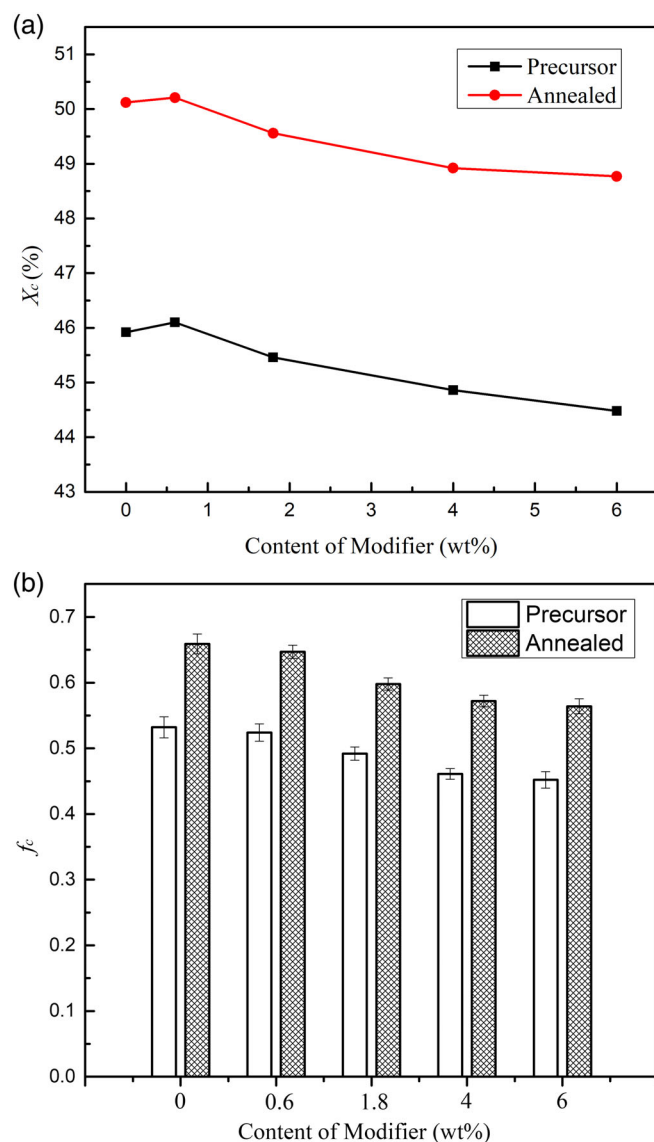


Figure 8. (a) X_c and (b) f_c of PP-PP-g-PAGK-SiO₂ precursor and annealed films with different PP-g-PAGK contents. [Color figure can be viewed at wileyonlinelibrary.com]

maximum value of 51.6% when the coupling agent content was 0.6 wt %. With an increase in the coupling agent content, the orientation of the lamellar structures in the precursor film was destroyed, and the porosity decreased.

The microscopic morphologies of the microporous membranes with different PP-g-PAGK contents were observed by SEM, and the results are shown in Figure 10. Table III shows the results of P_s , l_b , and the number of pores in the microporous membranes as analyzed by Image-Pro Plus. Compared to that of the microporous membrane without a coupling agent [Figure 10(a)], it was obvious that the main lamellar layer of the membrane containing 0.6 wt % PP-g-PAGK [Figure 10(b)] was more open. Furthermore, the number of pores increased; the pore size was more uniform. These results could also be inferred from Table III. The porosity of the microporous membranes without PP-g-PAGK was 12.17%, but it increased to 17.75% after the addition of

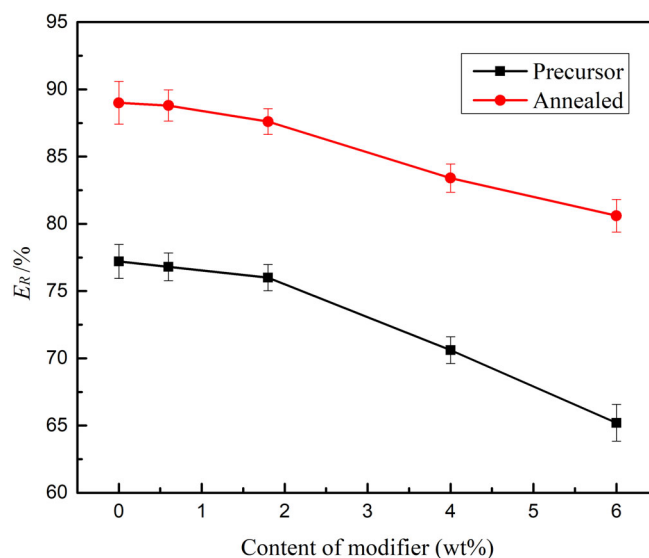


Figure 9. E_R values of PP-PP-g-PAGK-SiO₂ precursor and annealed films with different PP-g-PAGK contents. [Color figure can be viewed at wileyonlinelibrary.com]

0.6 wt % PP-g-PAGK. The number of pores and l_b also increased significantly. When the content of the macromolecular coupling agent was higher than 0.6 wt %, P_s decreased; this was consistent with the results shown in Table II (imbibition method).

The hydrophilicity of the microporous membranes was characterized in terms of the WVTR and water contact angle. Figure 11 shows the relationship between the WVTR and water contact angle of microporous membranes with different PP-g-PAGK contents. PP-g-PAGK was slightly polar in nature, and the number of polar groups on the surface of the microporous membrane increased gradually with increasing content of PP-g-PAGK, because of which the water contact angle of the microporous membranes decreased from 91.6 to 82.5°.

As shown in Figure 11, when the content of PP-g-PAGK was 0.6 wt %, the WVTR of the microporous membrane increased from the original 2936 g m⁻²/24 h to a maximum of 3360 g m⁻²/24 h. At this PP-g-PAGK content, the obtained microporous membrane exhibited the best permeability and porosity; further, the perfect pore structure facilitated water vapor transmission. Therefore, the highest WVTR was obtained for this membrane. With a further increase in the content of PP-g-PAGK, the orientation of the lamellar structure in the precursor film was destroyed along with the pore structure of the microporous membrane. The WVTR of the microporous membranes exhibited a decreasing trend, which was a joint effect of the pore structure and hydrophilicity.

Table II. Gurley Values and Porosities of PP-PP-g-PAGK-SiO₂ Microporous Membranes with Different PP-g-PAGK Contents (SiO₂ Content = 2 wt %)

PP-g-PAGK content (wt %)	0	0.6	1.8	4	6
Gurley value (s)	250	239	242	248	251
Porosity (%)	50.8	51.6	51.3	50.9	50.7

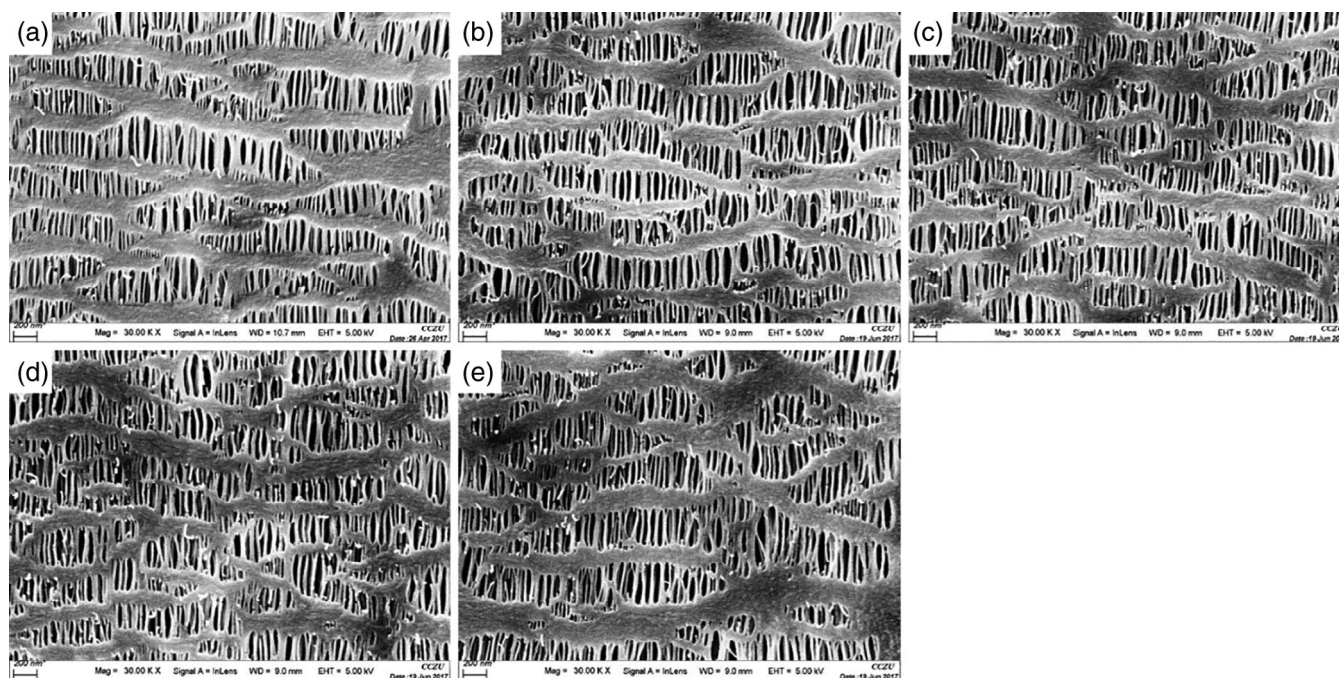


Figure 10. SEM images of PP-PP-g-PAGK-SiO₂ microporous membranes with 2 wt % SiO₂ and different contents of PP-g-PAGK: (a) 0, (b) 0.6, (c) 1.8, (d) 4, and (e) 6 wt %.

Figure 12 illustrates the thermal shrinkage rate of microporous membranes with different macromolecular coupling agent contents and pure PP microporous membranes at different temperatures. To simulate the heating process in lithium batteries, the tests were conducted at 110, 120, 130, 140, and 150°C. We observed that when the PP-g-PAGK content was less than 4 wt %, the heat resistance of the microporous membranes was higher than that of the sample without PP-g-PAGK. When the content of the coupling agent was 6 wt %, the heat shrinkage rate increased markedly, and the heat resistance of the microporous membrane decreased. A possible explanation is as follows. Some SiO₂ aggregates forced PP chains around stretching, and these chains curled up to equilibrium conformation when heated. More SiO₂ aggregates led to higher shrinkage. The coupling action of the PP-g-PAGK macromolecule induced an even dispersion of SiO₂ in PP; this was good for the crystallization of PP and reduced the heat shrinkage. On the other hand, excess PP-g-PAGK reduced the X_c of the precursor membrane, promoted PP segment motion via a plasticizing effect, and gave birth to the increase in the heat shrinkage. When the addition of PP-g-PAGK was 0.6 wt %, the heat resistance achieved the best result.

Table III. P_s , l_b , and Pore Number (N_h) Values of PP-PP-g-PAGK-SiO₂ Microporous Membranes with Different PP-g-PAGK Contents (SiO₂ Content = 2 wt %)

PP-g-PAGK content (wt %)	0	0.6	1.8	4	6
P_s (%)	12.17	17.75	15.62	13.91	13.99
l_b (nm)	49.2	54.7	54.0	52.8	53.0
N_h	642	712	690	639	604

Cycling Performance of Lithium-Ion Cells Prepared with PP-PP-g-PAGK-SiO₂ Compound Microporous Membranes

A pure PP microporous membrane and a composite microporous membrane containing 0.6 wt % PP-g-PAGK were used to prepare 2025 coin cells. The cycling performance of the battery was tested with a Neware system.

Figure 13 compares the cycle performance of coin cells assembled with different separators as a function of the cycle number. The initial specific discharge capacities of the two cells were 108 and 114 mAh/g, respectively, but the discharge capacities of the two cells decreased to 97 and 103 mAh/g, respectively, with increasing

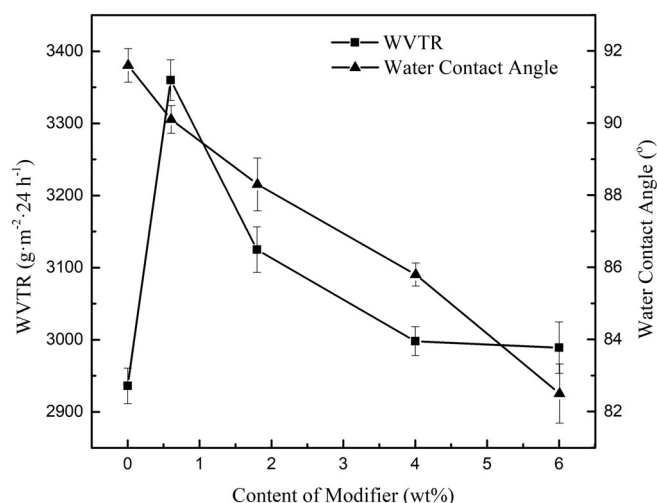


Figure 11. WVTRs and water contact angles of PP-PP-g-PAGK-SiO₂ microporous membranes with different PP-g-PAGK contents (SiO₂ content = 2 wt %).

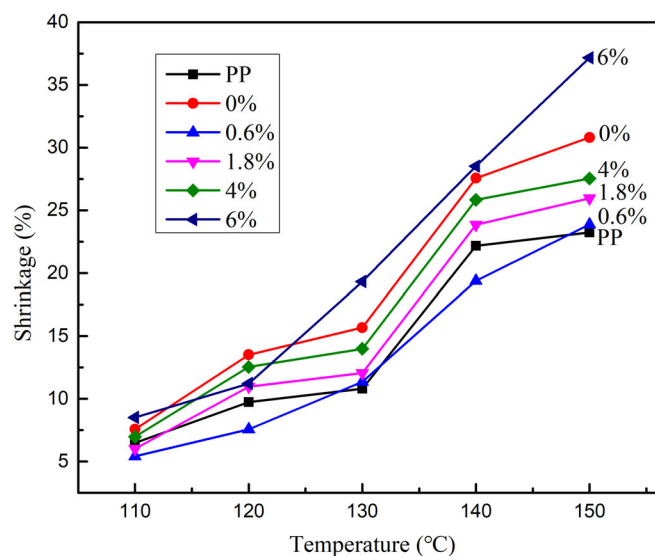


Figure 12. Thermal shrinkage rates of PP-PP-g-PAGK-SiO₂ microporous membranes with different contents of PP-g-PAGK and pure PP microporous membranes (SiO₂ content = 2 wt %). [Color figure can be viewed at wileyonlinelibrary.com]

cycle number. The respective capacity attenuation rates of the cells were 10.2 and 9.6%. In terms of capacity attenuation, the cell made of the composite microporous membrane was partially upgraded, because of which the cell stability was better.

The discharge capacities of the cells fabricated with different separators are compared in Figure 14. The discharge capacity of the cell assembled with the PP-PP-g-PAGK-SiO₂ compound microporous membrane was higher at all C rates. The difference in the discharge specific capacity at 0.1 C was about 6–10 mAh/g, whereas the difference between the two cells increased to 20–25 mAh/g at 10 C. With an increase in the discharge rate, the difference between the discharge capacities of the two cells increased; this was caused by an increase in the porosity of the composite

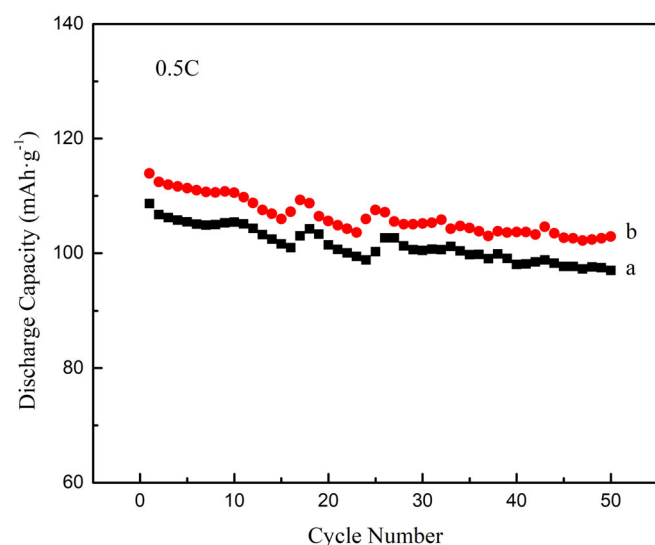


Figure 13. Cycle performance of coin cells assembled with the (a) PP microporous membrane and (b) PP-PP-g-PAGK-SiO₂ compound microporous membrane. [Color figure can be viewed at wileyonlinelibrary.com]

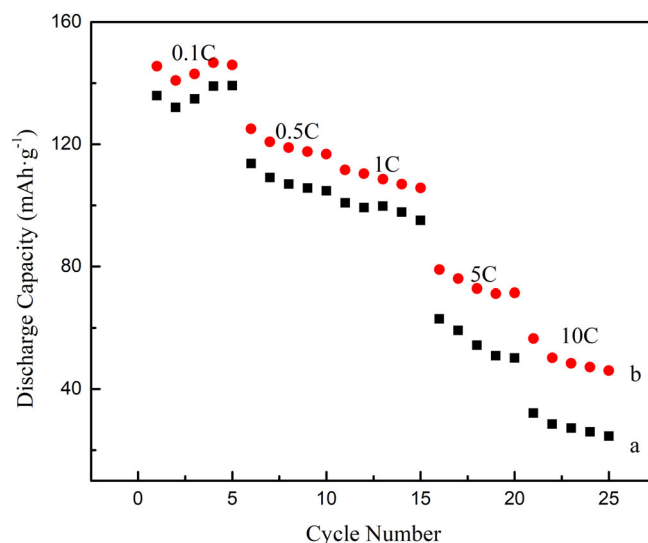


Figure 14. C-rate capability of coin cells assembled with the (a) PP microporous membrane and (b) PP-PP-g-PAGK-SiO₂ compound microporous membrane. [Color figure can be viewed at wileyonlinelibrary.com]

microporous membrane and its polarity. Thus, the multiplier cycle performance of the battery made with the composite microporous membrane was improved.

CONCLUSIONS

To improve the hydrophilicity and heat resistance of the microporous membrane, SiO₂ particles were introduced by melt blending. Although the permeability, porosity, and hydrophilicity of the microporous membranes reached their maximum values at an SiO₂ content of 2 wt %, its heat resistance decreased slightly. A macromolecular coupling agent, PP-g-PAGK, was prepared successfully and introduced into the PP-SiO₂ system to improve the dispersion of the SiO₂ particles in PP. With the addition of PP-g-PAGK, the segments of KH570 reacted with —OH on the surface of silica, and the dispersion of silica in PP was improved; this was good for the crystallization of PP and reduced the heat shrinkage. On the other hand, excess PP-g-PAGK reduced the X_c of the precursor membrane, promoted PP segmental motion, and gave birth to the increase in the heat shrinkage. When PP-g-PAGK content was 0.6 wt %, the heat resistance of the PP-PP-g-PAGK-SiO₂ compound microporous membranes was best and was slightly higher than that of the pure PP microporous membrane. Additionally, the cycle performance of the cell assembled with the PP-PP-g-PAGK-SiO₂ compound microporous membrane was better than that of the cell constructed with the pure PP membrane.

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